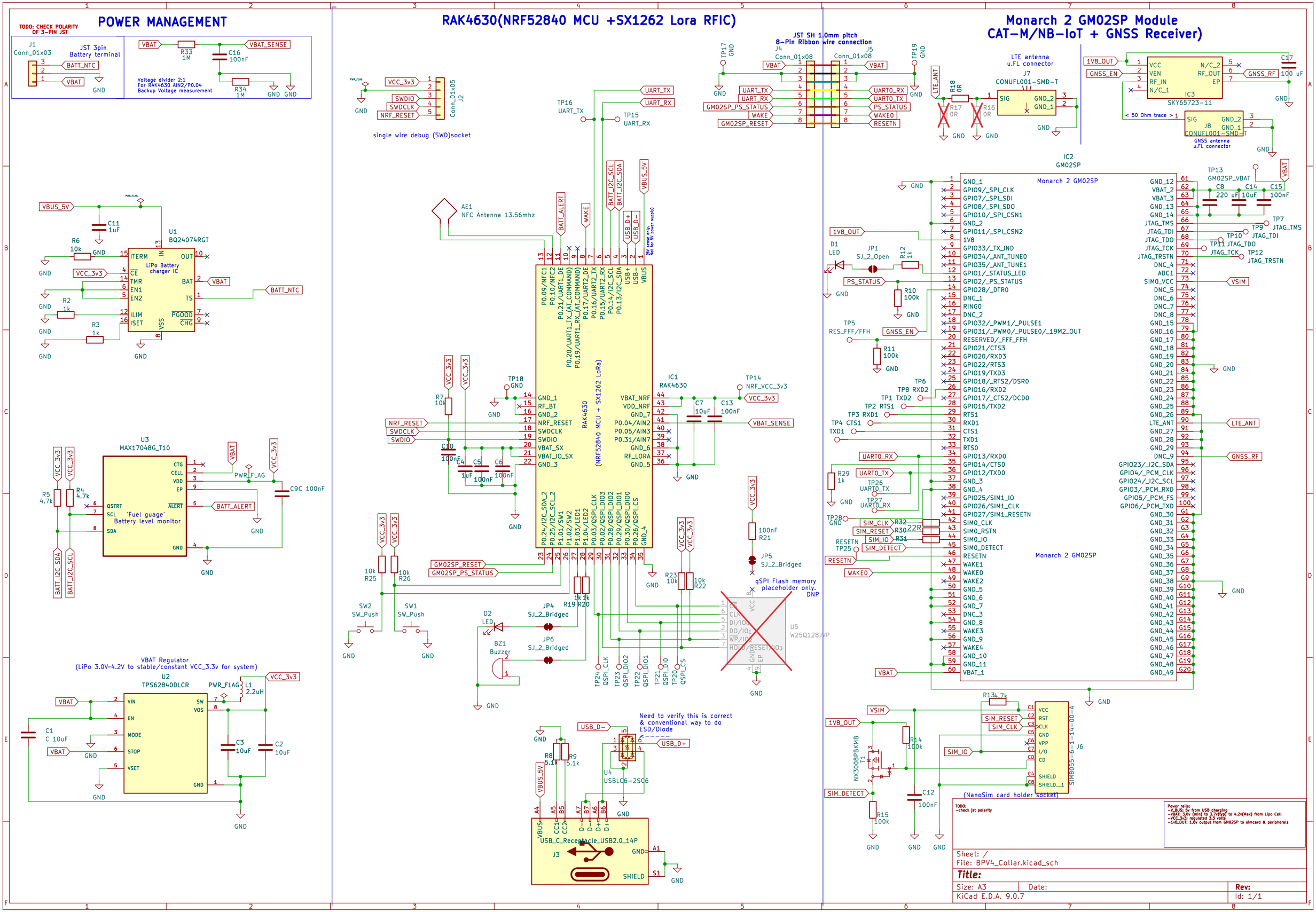
POWER MANAGEMENT
TODO: CHECK POLARITY OF 3-PIN JST
J1 Conn_01x03
JST 3pin Battery terminal
BATT_NTC
VBAT
GND
R33 1M
C16 100nF
VBAT_SENSE
Voltage divider 2:1
For RAK4630 AIN2/P0.04
Backup Voltage measurement
R34 1M
U1 BQ24074RGT
LiPo Battery charger IC
ITERM
CE
TMR
EN1
EN2
ILIM
ISET
IN
OUT
BAT
TS
PGOOD
CHG
VSS
VBUS_5V
C11 1uF
R6 10k
VCC_3v3
R2 1k
R3 1k
BATT_NTC
U3 MAX17048G_T10
'Fuel guage'
Battery level monitor
CTG
CELL
VDD
EP
QSTRT
SCL
SDA
ALERT
GND
R5 4.7k
R4 4.7k
BATT_I2C_SDA
BATT_I2C_SCL
BATT_ALERT
C9C 100nF
PWR_FLAG
VBAT Regulator
(LiPo 3.0V-4.2V to stable/constant VCC_3.3v for system)
U2 TPS62840DLCR
VIN
EN
MODE
STOP
VSET
SW
VOS
L1 2.2uH
C1 C 10uF
C3 10uF
C2 10uF
RAK4630(NRF52840 MCU +SX1262 Lora RFIC)
single wire debug (SWD)socket
J2 Conn_01x05
SWDIO
SWDCLK
NRF_RESET
TP16 UART_TX
TP15 UART_RX
AE1 NFC Antenna 13.56mhz
BATT_ALERT
WAKE
BATT_I2C_SCL
BATT_I2C_SDA
VBUS_5V
USB_D+
USB_D-
(5V sense only. Not for 5V power supply)
IC1 RAK4630
(NRF52840 MCU + SX1262 LoRa)
VBAT_NRF
VDD_NRF
GND_7
P0.04/AIN2
P0.05/AIN3
P0.31/AIN7
GND_6
RF_LORA
GND_5
TP14 NRF_VCC_3v3
C7 10uF
C13 100nF
VBAT_SENSE
R7 10k
TP18 GND
C10 100nF
C4 1uF
C5 100nF
C6 100nF
GM02SP_RESET
GM02SP_PS_STATUS
R25 10k
R26 10k
SW2 SW_Push
SW1 SW_Push
D2 LED
JP4 SJ_2_Bridged
JP6 SJ_2_Bridged
R19 1k
R20 1k
BZ1 Buzzer
R23 10k
R22 10k
R21 100nF
JP5 SJ_2_Bridged
qSPI Flash memory placeholder only. DNP
U5 W25Q128JVP
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
GND2
EP
TP24 QSPI_CLK
TP23 QSPI_DIO2
TP22 QSPI_DIO1
TP21 QSPI_DIO
TP20 QSPI_CS
Need to verify this is correct & conventional way to do ESD/Diode
USB_D-
USB_D+
R8 5.1k
R9 5.1k
U4 USBLC6-2SC6
USB_C_Receptacle_USB2.0_14P
J3
VBUS
CC1
CC2
D-
D+
SHIELD
A1
S1
Monarch 2 GM02SP Module
CAT-M/NB-IoT + GNSS Receiver)
JST SH 1.0mm pitch
8-Pin Ribbon wire connection
J4 Conn_01x08
J5 Conn_01x08
UART_TX
UART_RX
GM02SP_PS_STATUS
WAKE
GM02SP_RESET
UART0_RX
UART0_TX
PS_STATUS
WAKE0
RESETN
TP17 GND
TP19 GND
LTE antenna u.FL connector
J7 CONUFL001-SMD-T
R18 0R
R17 0R
R16 0R
LTE_ANT
1V8_OUT
GNSS_EN
VCC
VEN
RF_IN
N/C_1
N/C_2
RF_OUT
EP
IC3 SKY65723-11
GNSS_RF
C17 100 uF
< 50 Ohm trace >
J8 CONUFL001-SMD-T
GNSS antenna u.FL connector
IC2 GM02SP
Monarch 2 GM02SP
TP13 GM02SP_VBAT
C8 220 uF
C14 10uF
C15 100nF
TP7 JTAG_TMS
TP9 JTAG_TDI
TP10 JTAG_TDO
TP11 JTAG_TCK
TP12 JTAG_TRSTN
VSIM
LTE_ANT
GNSS_RF
D1 LED
JP1 SJ_2_Open
R12 1k
R10 100k
TP5 RES_FFF/FFH
R11 100k
TP6
TP8 RXD2
TP1 TXD2
TP2 RTS1
TP3 RXD1
TP4 CTS1
TXD1
R29 1k
TP26 UART0_TX
TP27 UART0_RX
TP28
SIM_CLK
SIM_RESET
SIM_IO
SIM_DETECT
R32 22R
R30
R31
RESETN
TP25
WAKE0
VBAT
R13 4.7k
SIM_RESET
SIM_CLK
SIM_IO
SIM_DETECT
R14 100k
C12 100nF
R15 100k
NX3008PBKMB
T1
VCC
RST
CLK
GND
VPP
I/O
CD
SHIELD
SHIELD_1
SIM8055-6-1-14-00-A
J6
(NanoSim card holder socket)
TODO:
-check jst polarity
Power rails:
-V_BUS: 5v from USB charging
-VBAT: 3.0v (min) to 3.7v(typ) to 4.2v(Max) from Lipo Cell
-VCC_3v3: regulated 3.3 volts
-1v8_OUT: 1.8v output from GM02SP to simcard & peripherals
Sheet: /
File: BPV4_Collar.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 1/1